

The empirical recurrence for OEIS A260474, recovered from its tail

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Abstract

OEIS A260474 records “Empirical recurrence of order 65” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 186$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 12$, so the recurrence is proved for every $n \geq 77$. Its last coefficient is $c_{65} = -79626240 \neq 0$, so no further tail satisfies anything shorter; any order-65 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A260474 is “Number of $(n+2) \times (7+2)$ 0..1 arrays with each 3×3 subblock having clockwise perimeter pattern 00000000 00000001 or 00010101.”. It has offset 1 and begins

1716, 5891, 36254, 161176, 839073, 4771920, 24138984, 123310754, ...

The entry states:

Empirical recurrence of order 65 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 12:07:28, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 186$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 186$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 65: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 12$ is the first whose minimal order is exactly the stated 65.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 372$ exact terms from $n = 12$ on, it returns order 65 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{65} c_i a(n-i),$$

c_1	1	c_{18}	−11901331	c_{35}	−3185369523	c_{52}	−80318525920
c_2	3	c_{19}	−9844013	c_{36}	−2455747541	c_{53}	−32040544504
c_3	112	c_{20}	−1859663	c_{37}	4144876615	c_{54}	−17202352512
c_4	101	c_{21}	37682958	c_{38}	8379615280	c_{55}	42864562368
c_5	−58	c_{22}	23365387	c_{39}	7019928802	c_{56}	10798818104
c_6	−3612	c_{23}	−17651888	c_{40}	−15474946399	c_{57}	2849863552
c_7	−5370	c_{24}	−108345322	c_{41}	−18449623671	c_{58}	−16117368384
c_8	−1038	c_{25}	−65964453	c_{42}	−14142836389	c_{59}	−3217168896
c_9	60962	c_{26}	106386772	c_{43}	40415567352	c_{60}	871195200
c_{10}	85524	c_{27}	254358930	c_{44}	33218964804	c_{61}	4541822208
c_{11}	29993	c_{28}	174077671	c_{45}	20621015652	c_{62}	1085630976
c_{12}	−548351	c_{29}	−362832010	c_{46}	−77707757434	c_{63}	48909312
c_{13}	−706200	c_{30}	−485201919	c_{47}	−48235198388	c_{64}	−443395584
c_{14}	−349962	c_{31}	−212478308	c_{48}	−25968898266	c_{65}	−79626240
c_{15}	3080980	c_{32}	1089235715	c_{49}	99111105080		
c_{16}	3296289	c_{33}	942550161	c_{50}	51573071168		
c_{17}	1750909	c_{34}	−476993917	c_{51}	26659825592		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 77$, and it is the recurrence OEIS A260474 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 12$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{65} - c_1 x^{64} - \dots - c_{65}$. Its constant term is $-c_{65} = 79626240$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination

of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 65 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 65, so $\deg q = 65$, and it is monic. It holds from some index t on. From $\max(t, 12)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 65, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 18 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 65, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -79626240 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-65 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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